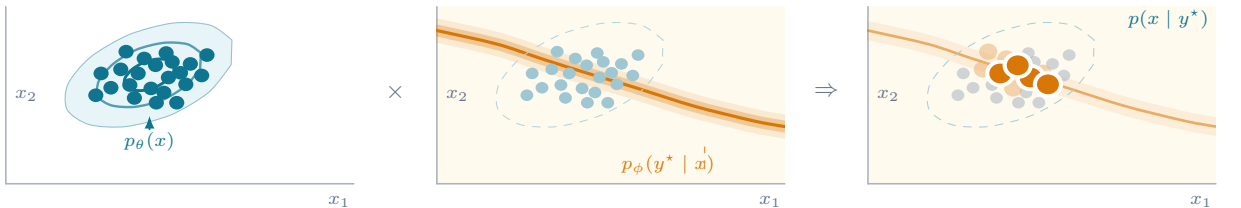
Bayes conditioning: $p(x \mid y^\star) \propto p_\phi(y^\star \mid x)\, p_\theta(x)$



Route A posterior known only up to Z_y: sample the product with MCMC

or fit $q_\psi(x)$ by variational inference, minimising $\mathrm{KL}(q_\psi \,\|\, p(\cdot \mid y^\star))$

Route B train on paired (x, y) data — no normaliser is ever formed → conditional generator learns $p_\theta(x \mid y)$ end to end

both target $p(x \mid y^\star)$